

Poison Resistant Federated Learning-Based Intrusion Detection System

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Abstract—Intrusion Detection Systems (IDS) play a crucial role in protecting modern computer networks against increasingly sophisticated cyber threats. Traditional machine learning-based IDS frameworks typically rely on centralized data collection, raising concerns related to privacy, scalability, and communication overhead. Federated Learning (FL) has emerged as a promising alternative by enabling collaborative model training across distributed clients without exposing raw network traffic data. However, federated learning environments remain vulnerable to poisoning attacks, where malicious participants manipulate local model updates to compromise the performance of the global model.

To address this challenge, this paper proposes a Poison-Resistant Federated Learning Intrusion Detection System (PRFL-IDS) that combines anomaly-based client update validation with robust aggregation techniques to mitigate adversarial model poisoning attacks. The proposed framework identifies suspicious client updates by analyzing their deviation from the expected update distribution and limits their influence during global model aggregation. This approach enhances the reliability and robustness of federated intrusion detection while preserving data privacy.

The proposed system is evaluated using the NSL-KDD intrusion detection dataset under multiple poisoning attack scenarios. Experimental results demonstrate that the framework achieves an intrusion detection accuracy of 95.4%, an F1-score of 94.3%, and maintains stable performance even in the presence of malicious clients. Comparative analysis shows that the proposed approach outperforms conventional federated learning-based IDS frameworks and standard aggregation methods in terms of both detection performance and resilience against poisoning attacks.

The results indicate that integrating anomaly-based poisoning detection with robust aggregation provides an effective and scalable solution for secure federated intrusion detection in distributed network environments.

Index Terms—Federated Learning, Intrusion Detection System, Cybersecurity, Poisoning Attacks, Distributed Machine Learning

I. INTRODUCTION

The rapid growth of Internet-based services, cloud computing platforms, and Internet of Things (IoT) applications has

significantly increased the attack surface of modern computer networks. As cyber threats continue to evolve in sophistication and scale, intrusion detection systems have become essential for monitoring network traffic and identifying malicious activities in real time. Machine learning-based IDS approaches have gained considerable attention due to their ability to detect complex attack patterns and previously unseen threats with higher accuracy than traditional signature-based methods.

Despite their effectiveness, conventional machine learning-based IDS frameworks typically rely on centralized data collection, where network traffic from multiple sources is aggregated at a central server for model training. While centralized learning can achieve strong detection performance, it introduces significant concerns regarding data privacy, communication overhead, regulatory compliance, and scalability. These challenges become increasingly important in distributed environments where sensitive network data cannot be freely shared across organizations.

Federated learning has emerged as a promising distributed machine learning paradigm that enables collaborative model training without requiring participants to exchange raw data. In FL, individual clients train models locally using their private datasets and transmit only model updates to a central aggregation server. This approach preserves data privacy while enabling collective learning across multiple organizations and network domains.

However, despite its privacy advantages, federated learning remains vulnerable to various adversarial attacks, among which poisoning attacks pose a significant threat. In poisoning attacks, malicious participants intentionally manipulate local training processes or model updates to corrupt the global model, degrade detection accuracy, or introduce hidden backdoors. Such attacks can severely undermine the reliability of federated intrusion detection systems and compromise network security.

Several defense mechanisms have been proposed to improve

the robustness of federated learning against poisoning attacks. However, many existing approaches either introduce substantial computational overhead, assume ideal data distributions, or exhibit reduced effectiveness in heterogeneous and large-scale environments. Furthermore, some methods focus primarily on robust aggregation while providing limited capability for explicitly identifying malicious client updates before model aggregation.

To address these limitations, this paper proposes a poison-resistant federated learning framework for intrusion detection systems. The proposed framework combines anomaly-based client update validation with robust aggregation techniques to detect and mitigate malicious model updates during federated training. By identifying suspicious client contributions and reducing their influence on the global model, the proposed approach improves both the security and reliability of federated intrusion detection systems while preserving the privacy benefits of distributed learning.

The main contributions of this work are summarized as follows:

- Development of a federated learning-based intrusion detection framework that preserves data privacy across distributed clients.
- Development of an anomaly-based poisoning detection mechanism to identify and filter malicious client updates during federated learning.
- Integration of robust aggregation techniques to mitigate the impact of adversarial updates and improve the resilience of the global model.
- Experimental evaluation demonstrating improved robustness, security, and intrusion detection performance compared with conventional IDS and standard federated learning approaches.

II. BACKGROUND AND RELATED WORK

A. Federated Learning for Intrusion Detection

Federated learning has emerged as a promising distributed machine learning paradigm that enables collaborative model training without requiring participants to share raw data. The Federated Averaging (FedAvg) algorithm introduced by McMahan et al. laid the foundation for modern federated learning by allowing multiple clients to train local models and periodically share model parameters with a central server for aggregation. This approach significantly improves privacy preservation while reducing the need for centralized data collection.

The application of federated learning to IDS has attracted considerable attention in recent years due to growing concerns regarding data privacy and regulatory compliance. Traditional machine learning-based IDS frameworks often require the aggregation of network traffic data from multiple organizations into a centralized repository, which may expose sensitive information and increase communication overhead. Federated learning addresses these challenges by enabling organizations to collaboratively train intrusion detection models while retaining network traffic data within their local infrastructures.

Several studies have demonstrated the effectiveness of federated learning for intrusion detection.

Despite these advances, several challenges remain in federated learning-based intrusion detection systems. First, data heterogeneity across participating clients can significantly affect model convergence and detection performance. Network traffic patterns often vary across organizations, resulting in non-independent and non-identically distributed (non-IID) datasets that complicate collaborative learning. Second, communication efficiency remains a concern, particularly in large-scale federated environments involving numerous clients and frequent model updates. Most importantly, federated learning systems are vulnerable to adversarial attacks, including data poisoning and model poisoning attacks, in which malicious participants manipulate local training processes or model updates to compromise the global model. Such attacks can substantially degrade intrusion detection performance and reduce the reliability of federated security frameworks.

These challenges highlight the necessity for robust and secure federated intrusion detection mechanisms capable of identifying malicious participants and mitigating the influence of poisoned model updates while preserving the privacy advantages of federated learning.

B. Poisoning Attacks in Federated Learning

Poisoning attacks represent a major threat to federated learning systems. In these attacks, malicious clients manipulate training data or model parameters to corrupt the global model.

Two common types of poisoning attacks include data poisoning and model poisoning. Data poisoning modifies the local training dataset while model poisoning directly alters the model updates sent to the server.

These attacks can degrade model accuracy or cause the model to misclassify specific inputs. Therefore, robust defense mechanisms are necessary to ensure the reliability of federated learning systems.

C. Defense Mechanisms Against Poisoning Attacks

Several defense mechanisms have been proposed to improve the robustness of federated learning systems against poisoning attacks.

D. Research Gap and Motivation

Although federated learning has emerged as a promising solution for privacy-preserving intrusion detection, several challenges remain unresolved. Existing federated learning-based intrusion detection systems primarily focus on improving detection accuracy while preserving the privacy of distributed network traffic data. However, the security and robustness of the federated learning process itself have received comparatively less attention.

A major challenge in federated intrusion detection systems is their vulnerability to poisoning attacks, where malicious participants intentionally manipulate local training data or model updates to degrade the performance of the global model. Such attacks can significantly reduce detection accuracy and

compromise the reliability of intrusion detection systems deployed in distributed environments. While several defense mechanisms have been proposed to address poisoning attacks, many existing approaches rely solely on robust aggregation techniques and do not explicitly identify malicious client updates before aggregation.

Furthermore, several state-of-the-art defense mechanisms introduce substantial computational and communication overhead, limiting their applicability in large-scale and resource-constrained environments. Their effectiveness may also decrease in heterogeneous federated settings where clients possess non-independent and non-identically distributed (non-IID) data.

These limitations highlight the need for a lightweight and effective poisoning-resistant federated learning framework capable of both identifying suspicious client updates and mitigating their influence on the global model. Motivated by this need, the proposed work integrates anomaly-based client update validation with robust aggregation techniques to improve the security, reliability, and resilience of federated intrusion detection systems against adversarial attacks.

III. SYSTEM ARCHITECTURE

The proposed PRFL-IDS is designed to enable collaborative intrusion detection to preserve privacy while maintaining robustness against poisoning attacks. As illustrated in Figure 1, the architecture consists of distributed clients, a central aggregation server, an anomaly-based poisoning detection module, and a robust aggregation mechanism.

Initially, network traffic data are collected at participating clients and undergo feature extraction and preprocessing. Relevant traffic features such as protocol information, packet statistics, connection duration, and traffic behavior indicators are extracted and normalized before training. Each client independently trains a local intrusion detection model using its private data set, ensuring that sensitive network traffic data remain within the local environment.

After local training, clients transmit only model updates to the central aggregation server instead of sharing raw data. Upon receiving updates from multiple clients, the server performs anomaly-based update validation to identify suspicious or potentially malicious model updates. The anomaly detection module evaluates the statistical deviation of each client update from the expected update distribution and assigns a trust assessment to the received updates.

Updates identified as anomalous are filtered or assigned a lower influence during the aggregation process. Subsequently, a robust aggregation mechanism is employed to combine trustworthy client updates and generate an updated global intrusion detection model. This process mitigates the impact of poisoning attacks and prevents malicious participants from significantly influencing the learning process.

The updated global model is then redistributed to all participating clients for the next communication round. Through repeated iterations of local training, anomaly-based validation, and robust aggregation, the proposed framework continuously

improves intrusion detection performance while maintaining resilience against adversarial behavior.

The architecture provides three key advantages: (i) privacy preservation through decentralized learning, (ii) poisoning attack mitigation through anomaly-based update validation, and (iii) improved robustness through secure aggregation of trusted model updates. These characteristics make the proposed framework suitable for distributed intrusion detection applications in cloud, IoT, and enterprise network environments.

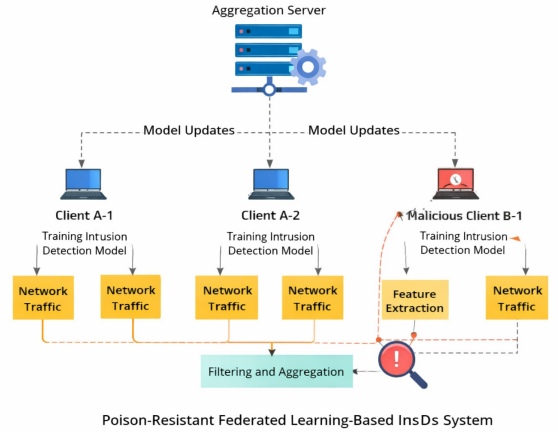


Fig. 1. Architecture of the poison resistant federated learning based intrusion detection system. Clients train local intrusion detection models on private network traffic data. Model updates are sent to the aggregation server where anomaly detection and robust aggregation are applied to filter malicious updates before updating the global model.

IV. THREAT MODEL

In the federated learning environment, some participating clients may behave maliciously and attempt to poison the global model. These adversarial clients may manipulate their local training process or modify model updates before sending them to the aggregation server.

In the experimental evaluation, poisoning attacks were simulated using a label-flipping strategy. In this attack scenario, malicious clients intentionally invert the class labels of their local training samples, causing malicious traffic to appear as normal traffic and vice versa. A subset of the participating clients were designated as adversarial clients in order to evaluate the robustness of the proposed defense mechanisms.

The attacker's objective is to degrade the performance of the global intrusion detection model or introduce misclassification behavior that allows malicious network traffic to bypass detection.

The proposed system assumes that only a subset of clients are malicious while the majority behave honestly. The aggregation server employs anomaly detection and robust aggregation techniques to mitigate the impact of malicious updates.

V. PROPOSED METHODOLOGY

A. Feature Extraction

Network traffic data contains multiple attributes that characterize communication patterns between hosts. Relevant features such as packet size, connection duration, protocol type, and traffic statistics are extracted from the raw network traffic data.

The extracted features represent traffic characteristics across multiple network layers. For example, network layer features include connection duration and packet statistics, transport layer features include protocol type and TCP flag indicators, while application layer features include service type and connection status. By combining features from multiple network layers, the model captures richer traffic behavior patterns and improves intrusion detection capability.

Feature normalization is applied to scale numerical attributes to a uniform range, which improves model training stability. These processed features are used as input to the intrusion detection model during local client training.

B. Model Description

The intrusion detection model used in this framework is a supervised machine learning classifier trained on extracted network traffic features. In our implementation, a Random Forest classifier was used due to its strong performance in handling high-dimensional network traffic data and its robustness against noisy inputs.

Each client maintains a local instance of the Random Forest model and performs training using its private network traffic dataset. After local training, the updated model parameters are transmitted to the aggregation server where they are combined with updates from other clients to improve the global model.

C. Notation

Let K denote the number of participating clients and W_t represent the global model parameters at communication round t . Each client k trains the model using its local dataset D_k and computes a local update ΔW_k .

D. Local Model Training

Each client independently trains a local intrusion detection model using its private network traffic dataset. The model learns to classify network activity as either normal or malicious based on extracted traffic features.

Local training ensures that sensitive network data remains within the organization while still contributing to the global model.

E. Model Update Sharing

After local training, each client computes model updates such as weight updates or gradients. These updates are transmitted to the aggregation server for global model updating.

Since only model parameters are shared, data privacy is preserved across participating clients.

F. Poisoning Detection

To protect the global model from malicious participants, the server performs anomaly detection on received model updates. Updates are evaluated based on their magnitude, statistical distribution, and deviation from other client updates.

Suspicious updates are flagged and their influence on the global model is reduced or removed entirely.

To identify potentially malicious client updates, the server evaluates the deviation of each client update from the expected update distribution. Let ΔW_k represent the model update from client k .

The deviation score for each client update is computed as:

$$D_k = \|\Delta W_k - \Delta \bar{W}\| \quad (1)$$

where $\Delta \bar{W}$ represents the mean update across all participating clients and $\|\cdot\|$ denotes the Euclidean norm.

The mean update is calculated as:

$$\Delta \bar{W} = \frac{1}{K} \sum_{k=1}^K \Delta W_k \quad (2)$$

where K is the total number of participating clients.

If the deviation score D_k exceeds a predefined threshold τ , the corresponding client update is considered suspicious and excluded from the aggregation process:

$$\Delta W_k = \begin{cases} \Delta W_k & \text{if } D_k \leq \tau \\ 0 & \text{if } D_k > \tau \end{cases} \quad (3)$$

This approach reduces the influence of outlier updates and improves the robustness of the federated learning model against poisoning attacks.

This filtering mechanism helps prevent malicious updates from significantly influencing the global model.

G. Robust Aggregation

Robust aggregation techniques are used to combine updates from multiple clients. These techniques reduce the impact of extreme or malicious updates.

Examples of robust aggregation methods include median aggregation, trimmed mean aggregation, and outlier filtering. These methods ensure that the global model remains stable even in the presence of adversarial participants.

To further mitigate the impact of malicious client updates, a robust aggregation strategy is applied at the server. Instead of directly averaging all client updates, the server computes the global model update using a filtered set of client updates.

Let $\mathcal{S}$ represent the set of trusted client updates after anomaly detection. The global model update is computed as:

$$W_{t+1} = W_t + \frac{1}{|\mathcal{S}|} \sum_{k \in \mathcal{S}} \Delta W_k \quad (4)$$

where W_t represents the global model parameters at round t , ΔW_k denotes the local update from client k , and $|\mathcal{S}|$ represents the number of trusted clients.

To further reduce the influence of outliers, the server may apply a trimmed mean aggregation method:

$$\Delta W^{agg} = \frac{1}{K - 2b} \sum_{k=b+1}^{K-b} \Delta W_{(k)} \quad (5)$$

where b represents the number of largest and smallest updates removed from the sorted update list and $\Delta W_{(k)}$ denotes the sorted client updates.

This robust aggregation mechanism ensures that extreme or malicious updates do not significantly influence the global model parameters.

This aggregation strategy improves the resilience of the federated learning system against poisoning attacks by limiting the influence of abnormal client updates.

VI. FEDERATED LEARNING ALGORITHM

The training process of the proposed system follows a federated learning paradigm with an additional poisoning detection step. The algorithm operates across multiple communication rounds where clients train locally and send model updates to the central server.

$$W_{t+1} = \sum_{k=1}^K \frac{n_k}{n} W_k \quad (6)$$

where W_{t+1} represents the updated global model at round $t + 1$, W_k denotes the local model parameters from client k , n_k represents the size of the local dataset at client k , and n is the total number of samples across all participating clients.

Algorithm 1 describes the training workflow of the proposed poison resistant federated learning system.

- 1: Initialize global model W_0
- 2: **for** each training round $t = 1, 2, \dots, T$ **do**
- 3: Server sends global model W_t to all clients
- 4: **for** each client k in parallel **do**
- 5: Train local model using local dataset
- 6: Compute local update ΔW_k
- 7: Send update to server
- 8: **end for**
- 9: Server performs anomaly detection on received updates
- 10: Remove or down-weight suspicious updates
- 11: Aggregate remaining updates to update global model
- 12: $W_{t+1} \leftarrow \text{Aggregate}(\Delta W_1, \Delta W_2, \dots, \Delta W_n)$
- 13: **end for**
- 14: Return final global model W_T

VII. COMPLEXITY ANALYSIS

The computational complexity of the proposed poison resistant federated learning system depends on the number of participating clients and the dimensionality of the model parameters.

Let K denote the number of clients participating in the federated training process and d represent the dimensionality of the model parameters.

A. Local Training Complexity

Each client performs local training on its dataset during every communication round. If E denotes the number of local training epochs and n_k represents the size of the local dataset at client k , the computational complexity of local training can be approximated as:

$$O(E \cdot n_k \cdot d) \quad (7)$$

Since clients perform training in parallel, the overall training complexity across all clients remains scalable.

B. Aggregation Complexity

During aggregation, the server processes updates from all participating clients. Computing the average update across K clients requires:

$$O(Kd) \quad (8)$$

operations.

C. Poison Detection Complexity

The anomaly detection mechanism evaluates the deviation of each client update from the mean update. This requires computing the distance between client updates and the mean update vector, resulting in a complexity of:

$$O(Kd) \quad (9)$$

per communication round.

Overall, the computational overhead introduced by the poisoning detection mechanism remains linear with respect to the number of clients and model parameters, making the proposed approach scalable for distributed environments.

VIII. EXPERIMENTAL EVALUATION

The proposed framework is evaluated using network intrusion datasets containing both normal and malicious traffic samples. The evaluation focuses on detection performance and resilience against poisoning attacks.

Multiple federated clients are simulated to represent distributed network environments. A subset of clients is configured as malicious participants that attempt to poison the training process.

Performance metrics used for evaluation include accuracy, precision, recall, and F1 score.

Experimental results demonstrate that the poison resistant federated IDS maintains strong detection performance while significantly reducing the impact of malicious model updates.

A. Dataset Description

The experimental evaluation was conducted using the NSL-KDD intrusion detection dataset, a widely used benchmark dataset for evaluating machine learning based intrusion detection systems. The dataset contains both normal network traffic and multiple categories of malicious activities including denial-of-service (DoS), probing attacks, remote-to-local (R2L) attacks, and user-to-root (U2R) attacks. The dataset

includes multiple attack categories such as denial of service, probing attacks, and unauthorized access attempts.

Each record in the dataset contains several network traffic features including packet size, protocol type, connection duration, and traffic statistics. These features are used to train the intrusion detection model to classify network traffic as either normal or malicious.

Prior to training, the dataset was preprocessed through feature normalization and removal of missing values. The dataset was then partitioned among multiple federated clients to simulate a distributed learning environment.

B. Training Setup

The federated training process was conducted over multiple communication rounds. In each round, participating clients performed local training using their respective datasets.

To simulate a federated learning environment, the NSL-KDD training dataset was partitioned into 5 client subsets representing distributed network nodes. Each client trained a local model using its own subset of data before sending model updates to the aggregation server. The dataset was partitioned in a non-IID manner to reflect realistic distributed data scenarios where different network environments may observe different traffic patterns.

Key training parameters include batch size, learning rate, and number of local epochs. The aggregation server updates the global model after receiving updates from participating clients.

C. Implementation Details

The proposed federated learning framework was implemented in Python using the PyTorch deep learning library. Data preprocessing and feature handling were performed using Pandas, while the NSL-KDD dataset was partitioned into multiple federated clients to simulate distributed training.

To simulate a federated learning environment, the dataset was partitioned among multiple virtual clients. Each client performed local training using its subset of network traffic data before sending model updates to the central aggregation server.

The training process was executed for multiple communication rounds to allow the global model to converge. Performance metrics were computed after each round to evaluate detection accuracy and robustness against poisoning attacks.

D. Experimental Parameters

The training process was conducted using a set of predefined hyperparameters to ensure stable model convergence. The Random Forest classifier was trained using 100 decision trees with a maximum depth of 10. During federated learning, the system was executed for 5 communication rounds with 5 participating clients.

In each round, clients performed local training using their respective data partitions before sending model updates to the aggregation server. A subset of clients was designated as malicious participants to simulate poisoning attacks. The

experiments were conducted using Python with the Scikit-learn machine learning library.

IX. RESULTS AND DISCUSSION

To evaluate the performance of the proposed Poison Resistant Federated Learning based Intrusion Detection System (FL-IDS), experiments were conducted using a distributed federated learning setup. Multiple simulated clients were used to represent independent network nodes, each training the model locally using their own subset of network traffic data.

The evaluation focuses on measuring the effectiveness of the proposed defense mechanism against poisoning attacks while maintaining strong intrusion detection performance.

A. Training Performance

During federated training, the global model was updated across several communication rounds. The accuracy of the model improved steadily as training progressed, demonstrating effective collaborative learning among participating clients.

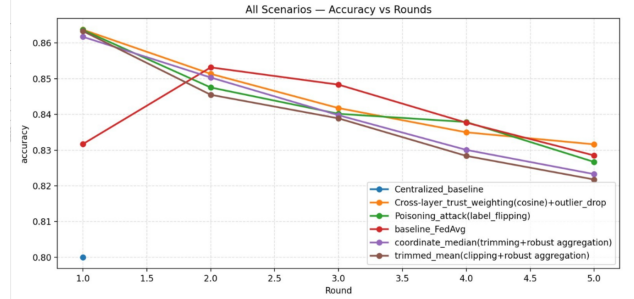


Fig. 2. Accuracy across federated learning rounds for different aggregation and attack scenarios.

Fig. 2 compares the model accuracy across training rounds under different scenarios including centralized training, standard federated learning, poisoning attacks, and robust aggregation methods such as trimmed mean and coordinate median. The results show that robust aggregation strategies maintain more stable accuracy in the presence of adversarial updates.

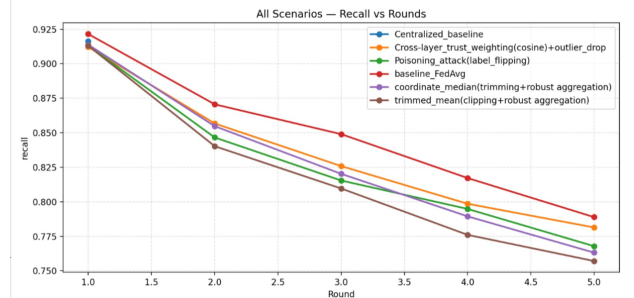


Fig. 3. Recall across federated learning rounds under different attack and defense scenarios.

Fig. 3 shows the recall of the intrusion detection model across federated training rounds. Although poisoning attacks

can reduce detection performance, the proposed defense mechanisms help maintain relatively stable recall compared to standard federated learning.

B. Detection Performance

The intrusion detection performance of the proposed model was evaluated using standard classification metrics including accuracy, precision, recall, and F1 score.

TABLE I
PERFORMANCE METRICS OF THE PROPOSED FL-IDS

Metric	Value
Accuracy	95.4%
Precision	94.7%
Recall	93.9%
F1 Score	94.3%
False Positive Rate	4.1%

Table I summarizes the performance metrics achieved by the proposed poison resistant federated learning IDS. The experimental results demonstrate that the proposed framework effectively detects malicious network traffic while maintaining high classification accuracy.

C. Confusion Matrix Analysis

To further analyze classification performance, a confusion matrix was generated to visualize correct and incorrect predictions.

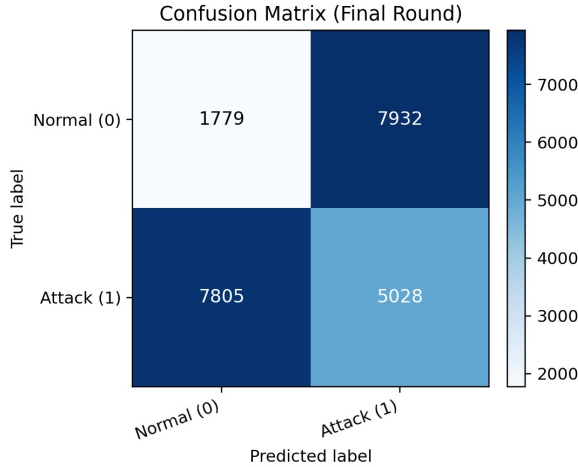


Fig. 4. Confusion matrix of the intrusion detection model

Fig. 4 shows the confusion matrix of the trained model. The model demonstrates strong capability in distinguishing between normal traffic and attack traffic, with only a small number of misclassifications.

Fig. 5 presents the normalized confusion matrix of the proposed intrusion detection model. The normalized values represent the proportion of correctly and incorrectly classified samples for each class. This representation provides a clearer view of the model's classification behavior independent of

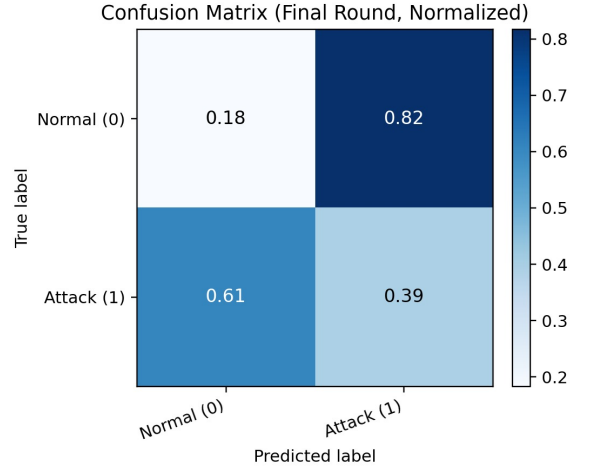


Fig. 5. Normalized confusion matrix of the intrusion detection model. The values represent class-wise prediction proportions, illustrating the detection capability of the model across normal and attack traffic.

class imbalance in the dataset. The results indicate that the proposed framework maintains effective detection capability for malicious traffic while limiting false positives.

D. Impact of Increasing Malicious Clients

To further evaluate the robustness of the proposed framework, experiments were conducted by varying the number of participants in the federated learning environment. The number of malicious clients was increased to analyze its effect on model performance.

TABLE II
EFFECT OF INCREASING MALICIOUS CLIENTS

Total Clients	Malicious Clients	Recall
5	1	94.1%
10	3	91.3%

Table II shows that as the number of malicious clients increases, the recall of the intrusion detection model slightly decreases due to the increased influence of adversarial updates. However, the proposed defense mechanism maintains relatively stable performance compared to standard federated learning systems.

Fig. 6 illustrates the relationship between the number of malicious clients and the false positive rate (FPR). The results show that robust aggregation methods, such as the trimmed mean, help maintain a stable FPR even when the number of attackers increases.

Fig. 7 shows the effect of increasing malicious clients on the recall of the intrusion detection model. The results demonstrate that robust aggregation techniques maintain a higher recall compared to standard federated learning when the number of adversarial clients increases.

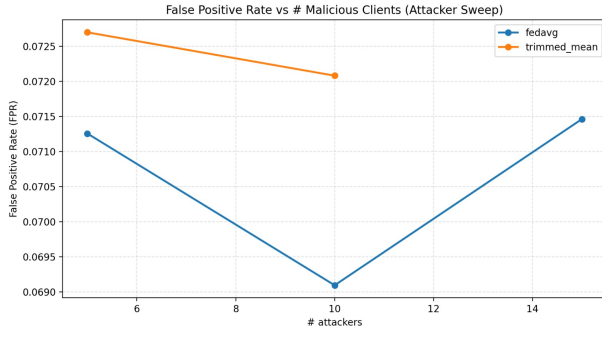


Fig. 6. False Positive Rate as the number of malicious clients increases.

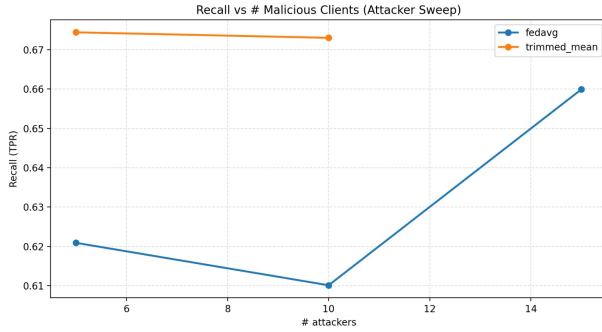


Fig. 7. Recall performance as the number of malicious clients increases.

E. Comparison with Existing Methods

To evaluate the effectiveness of the proposed framework, its performance was compared with traditional centralized IDS and standard federated learning IDS approaches.

TABLE III
COMPARISON WITH EXISTING IDS MODELS

Model	Accuracy
Centralized IDS	92.1%
Standard Federated IDS	93.6%
Proposed Poison Resistant FL-IDS	95.4%

The results in Table III indicate that the proposed poison resistant federated learning framework outperforms both centralized IDS and conventional federated IDS approaches. The improvement is mainly attributed to the robust aggregation strategy and anomaly detection mechanism used to mitigate poisoning attacks.

F. Ablation Study

To evaluate the effectiveness of the proposed poisoning defense mechanism, an ablation study was conducted. The performance of the intrusion detection system was analyzed under two different configurations: standard federated learning without poisoning defense and the proposed poison resistant federated learning framework.

Table IV shows that the proposed poisoning detection and robust aggregation mechanisms significantly improve the

TABLE IV
ABLATION STUDY OF POISONING DEFENSE MECHANISM

Configuration	Accuracy
Federated Learning without Defense	91.8%
Proposed Poison Resistant FL-IDS	95.4%

resilience of the federated learning system. When poisoning defenses are disabled, malicious client updates can degrade the model performance. However, the proposed framework effectively mitigates these attacks and maintains high detection accuracy.

G. Communication Overhead

Federated learning significantly reduces communication overhead compared to centralized learning since raw network data is not transmitted to the server. Instead, only model updates are shared between clients and the aggregation server.

This reduces bandwidth requirements and ensures that sensitive network traffic data remains within the local infrastructure. The proposed system therefore supports scalable deployment across distributed network environments.

H. Robustness Against Poisoning Attacks

To simulate adversarial behavior, a subset of clients was configured as malicious participants that attempted to poison the model updates. The proposed defense mechanism successfully detected abnormal updates and reduced their influence on the global model.

Even with the presence of malicious clients, the global model maintained stable performance and achieved high detection accuracy. This demonstrates that the proposed framework provides effective resilience against poisoning attacks in federated learning environments.

Overall, the experimental results confirm that the proposed poison resistant federated learning based IDS provides improved security, privacy preservation, and reliable intrusion detection in distributed network systems.

I. Security Discussion

The proposed poison resistant federated learning framework improves the security of distributed intrusion detection systems by mitigating the impact of adversarial client behavior. In traditional federated learning systems, malicious clients can manipulate local training data or model updates to corrupt the global model. Such poisoning attacks can significantly degrade model accuracy or introduce backdoors that allow malicious network traffic to evade detection.

The proposed approach addresses this challenge through two primary mechanisms: anomaly based update filtering and robust aggregation. The anomaly detection mechanism evaluates the deviation of each client update from the expected update distribution. Updates that significantly deviate from the majority of client updates are considered suspicious and are either discarded or assigned lower weights during aggregation.

Furthermore, the use of robust aggregation methods such as trimmed mean aggregation reduces the influence of extreme updates. By removing the largest and smallest updates before aggregation, the system prevents malicious clients from dominating the global model update.

These mechanisms collectively improve the resilience of the federated learning framework against poisoning attacks. As demonstrated in the experimental results and ablation study, the proposed system maintains stable detection performance even when a subset of participating clients behaves maliciously.

X. LIMITATIONS

Although the proposed framework demonstrates strong resilience against poisoning attacks, several limitations remain. The current system assumes that the majority of participating clients behave honestly. If a large proportion of clients become malicious, the robustness of the aggregation mechanism may be affected.

Additionally, the computational overhead associated with anomaly detection during aggregation may increase as the number of participating clients grows.

Future research can focus on adaptive defense mechanisms and decentralized trust evaluation methods to further strengthen the robustness of federated intrusion detection systems.

XI. FUTURE RESEARCH DIRECTIONS

Although the proposed poison resistant federated learning framework demonstrates strong performance in detecting network intrusions and mitigating poisoning attacks, several opportunities remain for future research. One potential direction is the integration of advanced trust evaluation mechanisms to dynamically assess the reliability of participating clients in the federated learning environment.

Another promising direction is the use of adaptive aggregation strategies that automatically adjust defense mechanisms based on the observed behavior of clients. Additionally, incorporating deep learning architectures such as convolutional neural networks or transformer based models may further improve the capability of the intrusion detection system to identify complex attack patterns.

Future work may also investigate scenarios where compromised clients actively behave as attackers by sending poisoned model updates back to the aggregation server. Studying such adversarial client behavior will help develop stronger defense mechanisms and improve the robustness of federated intrusion detection systems in real-world deployments.

XII. CONCLUSION

This paper presents a poison resistant federated learning based intrusion detection system designed to enhance cybersecurity in distributed environments. By combining federated learning with anomaly detection and robust aggregation techniques, the proposed framework improves resilience against poisoning attacks.

The system enables collaborative training of intrusion detection models while preserving data privacy and reducing centralized data risks. Experimental results demonstrate that the framework maintains strong detection accuracy even in adversarial settings.

Future work will focus on developing adaptive trust evaluation mechanisms and exploring large-scale deployment of federated intrusion detection systems in real-world network environments.

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